Narrative / Symbolic Space Analyzer

Characters in non-geocodable space: the kitchen, the field, the threshold — and WHO is in each.

What is non-geocodable (symbolic) space?

GIS_main maps geocodable space — real places that resolve to map coordinates (New York; Upson County, GA). But most of what happens in a narrative unfolds in non-geocodable space: kinds of place that carry social and symbolic meaning but have no latitude and longitude — the house, the field, the threshold of a door, the woods. "Kitchen versus field" is not a coordinate; it is a KIND of space. This tool takes the failed geocode and treats it as the SIGNAL, not the noise.

The ten space types

Every place word is classified into one of these. The names are what you will see on the axes of the charts and in the space_type column of the tables.

Space type	What it means	Words that classify to it
domestic_interior	Inside a home. The room, the private, the domestic — where the household is.	house, home, cottage, hut, cabin
field_labor	Worked land. The field, the farm, the plantation — where labour happens.	field, farm, farmyard, meadow, pasture
wild_forest	Uncultivated land. Woods, swamp, mountain — outside the settlement.	forest, wood, woods, woodland, wildwood
threshold_liminal	The in-between. Door, gate, porch, bridge, crossroads — crossed rather than occupied.	door, doorway, gate, gateway, threshold
royal_court	Seats of rule. Castle, palace, court.	castle, palace, court, throne, keep
sacred	Places of worship and the dead. Church, temple, graveyard.	church, chapel, cathedral, temple, altar
market_public	Where strangers meet by arrangement. Market, shop, street, town square.	market, marketplace, square, town, village
water_passage	Water and the ways along it. River, sea, harbour, road, railway.	river, stream, brook, creek, sea
subterranean	Below ground. Cave, cellar, mine, tunnel, dungeon, grave — enclosure, concealment, burial.	cellar, dungeon, basement, crypt, vault
tower_height	Above the ground. Tower, attic, roof, hill, balcony — vantage, isolation, elevation.	tower, turret, belfry, spire, attic

A place word the typology does not recognise is left unclassified and takes no part in the analyses.

How a word is classified

1. An exact lookup in lib/symbolic_space_typology.csv — a curated list you can edit.
2. Failing that, a WordNet hypernym climb: the word's ancestors are followed upwards, and if an anchor synset for a space type is on the path, that type is assigned. So "scullery" reaches domestic_interior without anybody listing it.

EDIT THE CSV to add your own domain's words — jail, courthouse, porch, cotton field. It is a plain three-column file (term, category, source) and the tool reads it at every run.

The twelve social-actor types

The tool also classifies WHO is acting. This is the attribute the STATIC analysis crosses against space, and it is derived for you — it used to be something you were told to code by hand.

Actor type	What it means	Words that classify to it
kin_family	Defined by family: mother, son, wife, widow.	mother, father, mom, dad, parent
child_youth	Defined by age: child, boy, pupil, orphan.	child, children, baby, infant, youth
laborer	Works with their hands, or for another: farmer, servant, sharecropper, slave.	farmer, laborer, labourer, worker, workman
professional	Trained occupation: doctor, lawyer, teacher, editor.	doctor, physician, surgeon, lawyer, attorney
authority_official	Holds civil office: judge, governor, mayor, coroner.	judge, governor, mayor, senator, congressman
military_police	Carries the state's force: sheriff, deputy, soldier, guard.	sheriff, deputy, policeman, constable, officer
clergy	Religious office: minister, priest, preacher, bishop.	minister, priest, preacher, pastor, reverend
merchant_trade	Buys and sells: merchant, shopkeeper, banker.	merchant, trader, shopkeeper, grocer, banker
elite_landowner	Owns land or rank: planter, landlord, master, lord.	planter, landowner, landlord, master, squire
crowd_collective	A group acting as one: mob, crowd, posse, jury, gang.	mob, crowd, throng, posse, gang
criminal_accused	Defined by the law against them: prisoner, convict, defendant, victim.	prisoner, convict, criminal, thief, murderer
generic_person	A person, but no social type: man, woman, people, someone.	

Same two passes: the curated list lib/social_actor_typology.csv first, then a WordNet hypernym climb, so "weaver" reaches laborer and "admiral" reaches military_police without being listed.

A PROPER NAME is never classified. WordNet does not know that Harry is a schoolboy, and guessing a person's social position from their name would be worse than saying nothing. For named characters, add the rows yourself — a corpus has few distinct characters however many documents it holds — or, for real historical people, see the note on knowledge graphs at the end.

The three steps

They run in order, and each one hands its output to the next.

- **1. BUILD — Extract characters in non-geocodable space.** reads a CoNLL table of your parsed corpus and finds every "actor in a place" event: a place word in a locative phrase, and the subject of the verb that governs it. It writes NLP_GIS_symbolic_actor_space_events_<corpus>.csv, one row per event, with the actor, the actor type, the space noun, the space type, the preposition and the sentence. That file is pushed straight into the INPUT box for the next two steps.
- **2. DYNAMIC — MAP characters moving.** traces how characters move between KINDS of space over the narrative (door → house → field → woods) and draws a directed movement graph. Set Location = space_noun and Sequence = Sentence ID — both are selected for you.
- **3. STATIC — Distribution of characters.** crosses the actor type against the space type: which kinds of people appear in which kinds of space, with a chi-square, Cramer's V and a heatmap of the over- and under-represented cells. Set Attribute = actor_type — selected for you — or your own column if you have coded one.

With a corpus of several documents, the movement is counted WITHIN each document: the end of one text is never joined to the start of the next.

Pronouns are dropped, and why that matters

An actor that is a pronoun — "he", "they", "which" — is left out. A pronoun is a reference to a character, not a character: kept in, pronouns become the largest "actors" in every table while naming nobody.

The BUILD step tells you how many events that cost. If the number is large, your corpus tells its story in pronouns, and the fix is COREFERENCE RESOLUTION: run it on the corpus (Parsers & annotators), parse the resolved text, and BUILD

from that CoNLL table. The pronouns become the people they stand for and those events come back with a name on them.

Adding an attribute of your own

actor_type says what kind of person somebody is. It does not say their gender, race or class. To cross one of those against space instead, add a column to the BUILD output and pick it as the Attribute column:

- **Gender:** the gender annotator can tag it for you.
- **Race, class:** there is no annotator. Code the list of DISTINCT actors once — a few hundred rows, not a few hundred thousand — and join it on the actor column.

A note on DBpedia and YAGO

For named REAL people, a knowledge graph can supply an occupation or a class. Measured against realistic corpora, though, the coverage is narrow: famous figures resolve (Mussolini → Politician, Du Bois → Scientist), fictional characters resolve only as "FictionalCharacter", and the local names that fill a historical newspaper corpus mostly do not resolve at all — and a common name can match the wrong entity entirely. Useful for a corpus of public figures; not a substitute for coding your own actors.

References

Moretti, F. (1998). Atlas of the European Novel, 1800–1900.

Lefebvre, H. (1974). The Production of Space.

Fellbaum, C., ed. (1998). WordNet: An Electronic Lexical Database. MIT Press.

See also the NLP Suite TIPS on GIS (Geographic Information System), Characters moving in time and space, and Extracting locations (NER).